Search-to-decision reduction for exponent-p class-two matrix-group isomorphism

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Editorial scope. Counted once: group search-to-decision Theorem A with its full A-prime tensor mechanism. Both runtime branches and every local support proof included; no separate count for gadgets, dynamic programming or direct applications. Preserves odd prime, class-two/exponent-p matrix-group model, finite-field isometry conventions and oracle bounds.

Allowed standard prerequisites (editorial).

- Finite-field linear algebra, standard permutation-group computations and imported FGS tensor gadget. Original definitions, complete monomial gadget and complete local search/dynamic-programming proofs retained. Separately cited [28] and [32] FGS background is retained with exact reference to Proposition 3.3. Complete source-local constructive Baer proof retained; full 74-item published bibliography.

Author notation and isometry models

Vector spaces. Let $\mathbb{F}$ be a field. In this paper we only consider finite-dimensional vector spaces over $\mathbb{F}$. We use $\mathbb{F}^n$ to denote the vector space of length-n column vectors. The ith standard basis vector of $\mathbb{F}^n$ is denoted $\vec{e}_i$. Depending on the context, 0 may denote the zero vector space, a zero vector, or an all-zero matrix. For S a set of vectors, we use $\langle S \rangle$ to denote the subspace spanned by elements in S.

Some groups. The general linear group of degree n over a field $\mathbb{F}$ is denoted by $\mathrm{GL}(n, \mathbb{F})$. The symmetric group of degree n is denoted by S_n. The natural embedding of S_n into $\mathrm{GL}(n, \mathbb{F})$ is to represent permutations by permutation matrices. The subgroup of $\mathrm{GL}(n, \mathbb{F})$ consisting of diagonal matrices is called the diagonal subgroup, denoted by $\mathrm{diag}(n, \mathbb{F})$. A monomial matrix is a product of a diagonal and a permutation matrix; equivalently, each row and each column has exactly one non-zero entry. The collection of monomial matrices forms a subgroup of $\mathrm{GL}(n, \mathbb{F})$, which we call the monomial subgroup and denote by $\mathrm{Mon}(n, \mathbb{F})$. It is the semi-direct product $\mathrm{diag}(n, \mathbb{F}) \rtimes S_n \cong (\mathbb{F}^*)^n \rtimes S_n$.

Matrices. Let $\mathrm{M}(\ell \times n, \mathbb{F})$ be the linear space of $\ell \times n$ matrices over $\mathbb{F}$, and $\mathrm{M}(n, \mathbb{F}) := \mathrm{M}(n \times n, \mathbb{F})$. Given $A \in \mathrm{M}(\ell \times n, \mathbb{F})$, A^t denotes the transpose of A. A matrix $A \in \mathrm{M}(n, \mathbb{F})$ is alternating, if for any $u \in \mathbb{F}^n$, $u^t A u = 0$. That is, A represents an alternating bilinear form. Note that in characteristic $\neq 2$, alternating is the same as skew-symmetric, but in characteristic 2 they differ (in characteristic 2, skew-symmetric=symmetric). The linear space of $n \times n$ alternating matrices over $\mathbb{F}$ is denoted by $\Lambda(n, \mathbb{F})$. The $n \times n$ identity matrix is denoted by I_n, and when n is clear from the context, we may just write I. The elementary matrix $E_{i,j}$ is the matrix with the (i,j)th entry being 1, and other entries being 0. The (i,j)-th elementary alternating matrix is the matrix $E_{i,j} - E_{j,i}$.

Matrix tuples. We use $\mathrm{M}(\ell \times n, \mathbb{F})^m$ to denote the linear space of m-tuples of $\ell \times n$ matrices. Boldface letters like $\mathbf{A}$ and $\mathbf{B}$ denote matrix tuples. Let $\mathbf{A} = (A_1, \ldots, A_m)$, $\mathbf{B} = (B_1, \ldots, B_m) \in \mathrm{M}(\ell \times n, \mathbb{F})^m$. Given $P \in \mathrm{M}(\ell, \mathbb{F})$ and $Q \in \mathrm{M}(n, \mathbb{F})$, $P\mathbf{A}Q := (PA_1Q, \ldots, PA_mQ) \in \mathrm{M}(\ell, \mathbb{F})$. Given $R = (r_{i,j})_{i,j \in [m]} \in \mathrm{M}(m, \mathbb{F})$, $\mathbf{A}^R := (A'_1, \ldots, A'_m) \in \mathrm{M}(m, \mathbb{F})$ where $A'_i = \sum_{j \in [m]} r_{j,i} A_j$.

Remark5. In particular, note that the coefficients in the formula of defining A'_i correspond to the entries in the ith column of R. While this choice is immaterial (we could have chosen the opposite convention), all of our later calculations are consistent with this convention.

Given $\mathbf{A}, \mathbf{B} \in \mathrm{M}(\ell \times n, \mathbb{F})^m$, we say that $\mathbf{A}$ and $\mathbf{B}$ are isometric, if there exists $P \in \mathrm{GL}(n, \mathbb{F})$, such that $P^t \mathbf{A} P = \mathbf{B}$. Finally, $\mathbf{A}$ and $\mathbf{B}$ are pseudo-isometric if there exist $P \in \mathrm{GL}(n, \mathbb{F})$ and $R \in \mathrm{GL}(m, \mathbb{F})$, such that $P^t \mathbf{A} P = \mathbf{B}^R$.

Matrix spaces. Linear subspaces of $\mathrm{M}(\ell \times n, \mathbb{F})$ are called matrix spaces. Calligraphic letters like $\mathcal{A}$ and $\mathcal{B}$ denote matrix spaces. By a slight abuse of notation, for $\mathbf{A} \in \mathrm{M}(\ell \times n, \mathbb{F})^m$, we use $\langle \mathbf{A} \rangle$ to denote the subspace spanned by those matrices in $\mathbf{A}$. For $\mathbf{A}, \mathbf{B} \in \mathrm{M}(n, \mathbb{F})^m$, we say that the spaces $\langle \mathbf{A} \rangle, \langle \mathbf{B} \rangle$ are isometric iff the tuples $\mathbf{A}, \mathbf{B}$ are pseudo-isometric.

Definition4 (The Alternating Matrix Space Isometry problem). Input: $A_1, \ldots, A_m$ and $B_1, \ldots, B_m$, $n \times n$ alternating matrices over a field $\mathbb{F}$. Decide: Is there a $L \in \mathrm{GL}(n, \mathbb{F})$, such that the linear span of $\{A_i : i \in [m]\}$ is equal to the linear span of $\{L^t B_i L : i \in [m]\}$?

3-way arrays. Let $\mathrm{T}(\ell \times n \times m, \mathbb{F})$ be the linear space of $\ell \times n \times m$ 3-way arrays over $\mathbb{F}$. We use the fixed-width teletypefont for 3-way arrays, like $\mathtt{A}, \mathtt{B}$, etc.. Given $\mathtt{A} \in \mathrm{T}(\ell \times n \times m, \mathbb{F})$, we can think of $\mathtt{A}$ as a 3-dimensional table, where the (i,j,k)th entry is denoted as $\mathtt{A}(i,j,k) \in \mathbb{F}$. We can slice $\mathtt{A}$ along one direction and obtain several matrices, which are then called slices. For example, slicing along the first coordinate, we obtain the horizontal slices, namely ℓ matrices $A_1, \ldots, A_\ell \in \mathrm{M}(n \times m, \mathbb{F})$, where $A_i(j,k) = \mathtt{A}(i,j,k)$. Similarly, we also obtain the lateral slices by slicing along the second coordinate, and the frontal slices by slicing along the third coordinate. We will often represent a 3-way array as a matrix whose entries are vectors. That is, given $\mathtt{A} \in \mathrm{T}(\ell \times n \times m, \mathbb{F})$, we can

write

$$\mathtt{A} = \begin{bmatrix} w_{1,1} & w_{1,2} & \cdots & w_{1,n} \\ w_{2,1} & w_{2,2} & \cdots & w_{2,n} \\ \vdots & \vdots & \ddots & \vdots \\ w_{\ell,1} & w_{\ell,2} & \cdots & w_{\ell,n} \end{bmatrix},$$

where $w_{i,j} \in \mathbb{F}^m$, so that $w_{i,j}(k) = \mathtt{A}(i,j,k)$. Note that, while $w_{i,j} \in \mathbb{F}^m$ are column vectors, in the above representation of $\mathtt{A}$, we should think of them as along the direction "orthogonal to the paper." Following [45], we call $w_{i,j}$ the tube fibers of $\mathtt{A}$. Similarly, we can have the row fibers $v_{i,k} \in \mathbb{F}^n$ such that $v_{i,k}(j) = \mathtt{A}(i,j,k)$, and the column fibers $u_{j,k} \in \mathbb{F}^\ell$ such that $u_{j,k}(i) = \mathtt{A}(i,j,k)$. Given $P \in \mathrm{M}(\ell, \mathbb{F})$ and $Q \in \mathrm{M}(n, \mathbb{F})$, let $P\mathtt{A}Q$ be the $\ell \times n \times m$ 3-way array whose kth frontal slice is PA_kQ. For $R = (r_{i,j}) \in \mathrm{GL}(m, \mathbb{F})$, let $\mathtt{A}^R$ be the $\ell \times n \times m$ 3-way array whose kth frontal slice is $\sum_{k' \in [m]} r_{k',k} A_{k'}$. Note that these notations are consistent with the notations for matrix tuples above, when we consider the matrix tuple $\mathbf{A} = (A_1, \ldots, A_k)$ of frontal slices of $\mathtt{A}$.

TheoremA. Let p be an odd prime, and let GpIso2Exp(p) denote the isomorphism problem for p-groups of class 2 and exponent p in the model of matrix groups over $\mathbb{F}_p$. For groups of order p^n, there is a search-to-decision reduction for GpIso2Exp(p) running in time $p^{O(n)} = \mathrm{poly}(|G|)$.

4 Search-to-decision reduction by restricting to monomial groups

4.1 The gadget restricting to the monomial group

In this section, we present the gadget that restricts to the monomial group in the setting of Alternating Matrix Space Isometry. To show this, we will need the concept of monomial isometry; see Some Groups above. Recall that a matrix is monomial if, equivalently, it can be written as DP where D is a nonsingular diagonal matrix and P is a permutation matrix. We say two matrix spaces $\mathcal{A}, \mathcal{B}$ are monomially isometric if there is some $M \in \mathrm{Mon}(n, \mathbb{F})$ such that $M^t \mathcal{A} M = \mathcal{B}$.

Lemma8. Alternating Matrix Space Monomial Isometry reduces to Alternating Matrix Space Isometry. More specifically, there is a $\mathrm{poly}(n, m)$-time algorithm r taking alternating matrix tuples to alternating matrix tuples, such that for $\mathbf{A}, \mathbf{B} \in \Lambda(n, \mathbb{F})^m$, the matrix spaces $\mathcal{A} = \langle \mathbf{A} \rangle$ and $\mathcal{B} = \langle \mathbf{B} \rangle$ are monomially isometric if and only if the matrix spaces $\langle r(\mathbf{A}) \rangle$ and $\langle r(\mathbf{B}) \rangle$ are isometric.

The gadget used in Lemma8 is essentially to apply the gadget in Proposition7 "in two directions." Still, to prove the correctness requires some work.

Proof. For $\mathbf{A} = (A_1, \ldots, A_m) \in \Lambda(n, \mathbb{F})^m$, define $r(\mathbf{A})$ to be the alternating matrix tuple $\widetilde{\mathbf{A}} = (\widetilde{A}_1, \ldots, \widetilde{A}_{m+n^2}) \in \Lambda(n + n^2, \mathbb{F})^{m+n^2}$, where

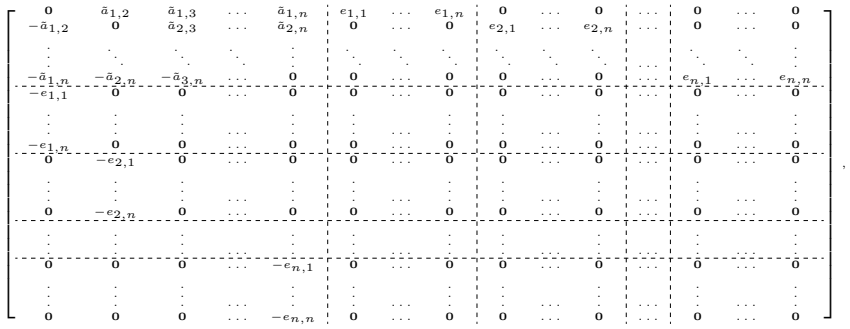
1. For $k = 1, \ldots, m$, $\widetilde{A}_k = \begin{bmatrix} A_k & 0 \\ 0 & 0 \end{bmatrix}$.
2. For $k = m + (i-1)n + j$, $i \in [n], j \in [n]$, $\widetilde{A}_k$ is the elementary alternating matrix $E_{i,in+j} - E_{in+j,i}$.

At this point, some readers may wish to look at the large matrix in Equation2 and/or at Figure2. It is clear that r can be computed in time $\widetilde{O}((m+n^2)(n^2+n)) = \text{poly}(n,m)$. Given alternating matrix tuples $\mathbf{A}, \mathbf{B}$, let $\mathcal{A}, \mathcal{B}$ be the corresponding matrix spaces they span, and let $\widetilde{\mathcal{A}} = \langle r(\mathbf{A}) \rangle$ and $\widetilde{\mathcal{B}} = \langle r(\mathbf{B}) \rangle$. We claim that $\mathcal{A}$ and $\mathcal{B}$ are monomially isometric if and only if $\widetilde{\mathcal{A}}$ and $\widetilde{\mathcal{B}}$ are isometric. To prove this, it will help to think of our matrix tuples $\mathbf{A}, \widetilde{\mathbf{A}}$, etc. as (corresponding to) 3-way arrays, and to view these 3-way arrays from two different directions. Towards this end, write the 3-way array corresponding to $\mathbf{A}$ as

$$\mathtt{A} = \begin{bmatrix} 0 & a_{1,2} & a_{1,3} & \cdots & a_{1,n} \\ -a_{1,2} & 0 & a_{2,3} & \cdots & a_{2,n} \\ -a_{1,3} & -a_{2,3} & 0 & \cdots & a_{3,n} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ -a_{1,n} & -a_{2,n} & -a_{3,n} & \cdots & 0 \end{bmatrix},$$

where $a_{i,j}$ are vectors in $\mathbb{F}^m$ ("coming out of the page"), namely $a_{i,j}(k) = A_k(i,j)$. The frontal slices of this array are precisely the matrices $A_1, \ldots, A_m$. The 3-way array corresponding to $\widetilde{\mathbf{A}} = r(\mathbf{A})$ is then the $(n+1)n \times (n+1)n \times (m+n^2)$ array:

$\widetilde{\mathtt{A}} =$



(2)

where $\widetilde{a}_{i,j} = \begin{bmatrix} a_{i,j} \\ 0 \end{bmatrix} \in \mathbb{F}^{m+n^2}$ (here think of the vector $a_{i,j}$ as a column vector, not coming out of the page; in the above array we then lay the column vector $\widetilde{a}_{i,j}$ "on its side" so that it is coming out of the page), and $e_{i,j} := e_{m+(i-1)n+j} \in \mathbb{F}^{m+n^2}$, which we can equivalently write as $\begin{bmatrix} 0^m \\ e_i \otimes e_j \end{bmatrix}$, where we think of $e_i \otimes e_j$ here as a vector of length n^2. Note that all the the nonzero blocks besides upper-left "$\mathtt{A}$" block only have nonzero entries that are strictly behind the nonzero entries in the upper-left block.

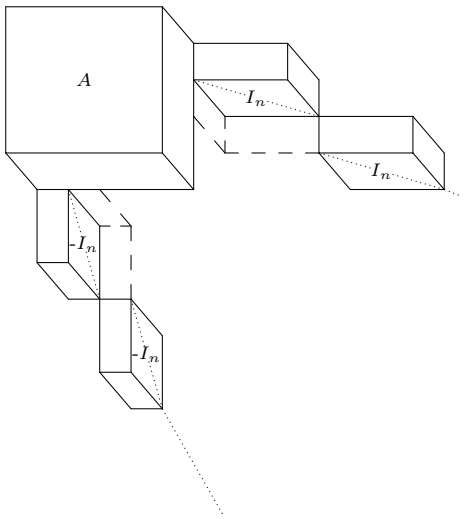


Figure2. Pictorial representation of the reduction for Lemma8. The second viewpoint, which we will also use below, is to consider the lateral slices of $\widetilde{\mathtt{A}}$, or equivalently, to view $\widetilde{\mathtt{A}}$ from the side. When viewing $\widetilde{\mathtt{A}}$ from the side, we see the $(n+1)n \times (m+n^2) \times (n+1)n$ 3-way array:

$$\mathtt{A}^{lat} = \left[\begin{array}{cccc:ccc:c:ccc} \ell_{1,1} & \ell_{1,2} & \dots & \ell_{1,m} & e_{n+1} & \dots & e_{2n} & \dots & 0 & \dots & 0 \\ \vdots & \ddots & \ddots & \vdots & \vdots & \ddots & \vdots & \ddots & \vdots & \ddots & \vdots \\ \ell_{n,1} & \ell_{n,2} & \dots & \ell_{n,m} & 0 & \dots & 0 & \dots & e_{n^2+1} & \dots & e_{n^2+n} \\ \hdashline 0 & 0 & \dots & 0 & e_1 & \dots & 0 & \dots & 0 & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots & \vdots & \ddots & \vdots & \dots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & 0 & 0 & \dots & e_1 & \dots & 0 & \dots & 0 \\ \hdashline \vdots & \ddots & \ddots & \vdots & \vdots & \ddots & \vdots & \ddots & \vdots & \ddots & \vdots \\ \hdashline 0 & 0 & \dots & 0 & 0 & \dots & 0 & \dots & e_n & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots & \vdots & \ddots & \vdots & \dots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & 0 & 0 & \dots & 0 & \dots & 0 & \dots & e_n \end{array}\right], \tag{3}$$

where every $\ell_{i,k} \in \mathbb{F}^{n^2+n}$ has only the first n components being possibly non-zero, namely, $\ell_{i,k}(j) = A_k(i,j)$ for $i \in [n], j \in [n], k \in [m]$ and $\ell_{i,k}(j) = 0$ for any $j > n$.

For the only if direction. Suppose there exist $P \in \mathrm{Mon}(n, \mathbb{F})$ and $Q \in \mathrm{GL}(m, \mathbb{F})$, such that $P^t \mathbf{A} P = \mathbf{B}^Q$. We can construct $\widetilde{P} \in \mathrm{Mon}(n+n^2, \mathbb{F})$ and $\widetilde{Q} \in \mathrm{GL}(m+n^2, \mathbb{F})$ such that $\widetilde{P}^t \widetilde{\mathbf{A}} \widetilde{P} = \widetilde{\mathbf{B}}^{\widetilde{Q}}$. In fact, we will show that we can take $\widetilde{P} = \begin{bmatrix} P & 0 \\ 0 & P' \end{bmatrix}$ where $P' \in \mathrm{Mon}(n^2, \mathbb{F})$, and $\widetilde{Q} = \begin{bmatrix} Q & 0 \\ 0 & Q' \end{bmatrix}$ where $Q' \in \mathrm{Mon}(n^2, \mathbb{F})$. It is not hard to see that this form already ensures that the first m matrices in the vector $\widetilde{P}^t \widetilde{\mathbf{A}} \widetilde{P}$ and those of $\widetilde{\mathbf{B}}^{\widetilde{Q}}$ are the same, since when $\widetilde{P}, \widetilde{Q}$ are of this form, those first m matrices are controlled entirely by the P (resp., Q) in the upper-left block of $\widetilde{P}$ (resp., $\widetilde{Q}$). The remaining question is then how to design appropriate P' and Q' to take care of the last n^2 matrices in these tuples. This actually boils down to applying the following simple identity, but

"in 3 dimensions:" Let P be the permutation matrix corresponding to $\sigma \in S_n$, so that $Pe_i = e_{\sigma(i)}$, and $e_i^t P = e_{\sigma^{-1}(i)}^t$. Let $D = \mathrm{diag}(\alpha_1, \ldots, \alpha_n)$ be a diagonal matrix. Then

$$P^t D P = \mathrm{diag}(\alpha_{\sigma^{-1}(1)}, \ldots, \alpha_{\sigma^{-1}(n)}). \tag{4}$$

To see how Equation4 helps in our setting, it is easier to focus attention on the lower right $n^2 \times n^2$ sub-array of $\mathtt{A}^{\mathrm{lat}}$, which can be represented as a symbolic matrix

$$M = \begin{bmatrix} x_1 I_n & 0 & \cdots & 0 \\ 0 & x_2 I_n & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & x_n I_n \end{bmatrix}.$$

Here we think of the x_i's as independent variables, whose indices correspond to "how far into the page" they are. That is, x_i corresponds to the vector $\vec{e}_i$ in $\mathtt{A}^{\mathrm{lat}}$, which is coming out of the page and has its only nonzero entry i slices back from the page. Then the action of P permutes the x_i's and multiplies them by some scalars, the action of P' is on the left-hand side, and the action of Q' is on the right-hand side. Let σ be the permutation supporting P. Then P sends M to

$$M^P = \begin{bmatrix} \alpha_{\sigma(1)} x_{\sigma(1)} I_n & 0 & \cdots & 0 \\ 0 & \alpha_{\sigma(2)} x_{\sigma(2)} I_n & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & \alpha_{\sigma(n)} x_{\sigma(n)} I_n \end{bmatrix}.$$

So setting $P' = \sigma \otimes I_n$, Q' the monomial matrix supported by $\sigma \otimes I_n$ with scalars being $1/\alpha_i$'s, we have $P'^t M^P Q' = M$ by Equation4.

For the if direction. Suppose there exist $\widetilde{P} \in \mathrm{GL}(n+n^2, \mathbb{F})$ and $\widetilde{Q} \in \mathrm{GL}(m+n^2, \mathbb{F})$, such that $\widetilde{P}^t \widetilde{\mathbf{A}} \widetilde{P} = \widetilde{\mathbf{B}}^{\widetilde{Q}}$. The key feature of these gadgets now comes into play: consider the lateral slices of $\widetilde{\mathtt{A}}$, which are the frontal slices of $\mathtt{A}^{\mathrm{lat}}$ (which may be easier to visualize by looking at Equation3 and Figure2). The first n lateral slices of $\widetilde{\mathtt{A}}$ and $\widetilde{\mathtt{B}}$ are of rank $\geq n$ and $< 2n$, while the other lateral slices are of rank $< n$ (in fact, they are of rank 1; note that without loss of generality we may assume $n > 1$, for the only 1×1 alternating matrix space is the zero space). Furthermore, left multiplying a lateral slice by $\widetilde{P}^t$ and right multiplying it by $\widetilde{Q}$ does not change its rank. However, the action of $\widetilde{P}$ here is by $\widetilde{P}^t \widetilde{\mathbf{A}} \widetilde{P}$, and while the $\widetilde{P}^t$ here corresponds to left multiplication on the lateral slices (=frontal slices of $\mathtt{A}^{\mathrm{lat}}$), the $\widetilde{P}$ on the right here corresponds to taking linear combinations of the lateral slices. In other words, just as $\mathtt{A}^{\mathrm{lat}}$ is the "side view" of $\widetilde{\mathtt{A}}$, $(\widetilde{P}^t \mathtt{A}^{\mathrm{lat}} \widetilde{Q})^{\widetilde{P}}$ is the side view of $(\widetilde{P}^t \widetilde{\mathtt{A}} \widetilde{P})^{\widetilde{Q}}$. Taking linear combinations of the lateral slices could, in principle, alter their rank; we will use the latter possibility to show that $\widetilde{P}$ must be of a constrained form. Write $\widetilde{P} = \begin{bmatrix} P_{1,1} & P_{1,2} \\ P_{2,1} & P_{2,2} \end{bmatrix}$ where $P_{1,1}$ is of size $n \times n$. We first claim that $P_{1,2} = 0$. For if not, then in $(\mathtt{A}^{\mathrm{lat}})^{\widetilde{P}}$ (the side view), one of the last n^2 frontal slices receives a nonzero contribution from one of the first n frontal slices of $\mathtt{A}^{\mathrm{lat}}$. Looking at the form of these slices from Equation3, we see that any such nonzero combination will have rank $\geq n$, but this is a contradiction since the corresponding slice in $\mathtt{B}^{\mathrm{lat}}$ has rank 1. Thus $P_{1,2} = 0$, and therefore $P_{1,1}$ must be invertible, since $\widetilde{P}$

is. Finally, we claim that $P_{1,1}$ has to be a monomial matrix. If not, then some frontal slice of $(\mathtt{A}^{\mathrm{lat}})^{\widetilde{P}}$ among the first n would have a contribution from more than one of these n slices. Considering the lower-right $n^2 \times n^2$ sub-matrix of such a slice, we see that it would have rank exactly kn for some $k \geq 2$, which is again a contradiction since the first n slices of $\mathtt{B}^{\mathrm{lat}}$ all have rank $< 2n$. It follows that $P_{1,1}^t A_i P_{1,1}$, $i \in [m]$, are in $\mathcal{B}$, and thus $\mathcal{A}$ and $\mathcal{B}$ are monomially isometric via $P_{1,1}$. □

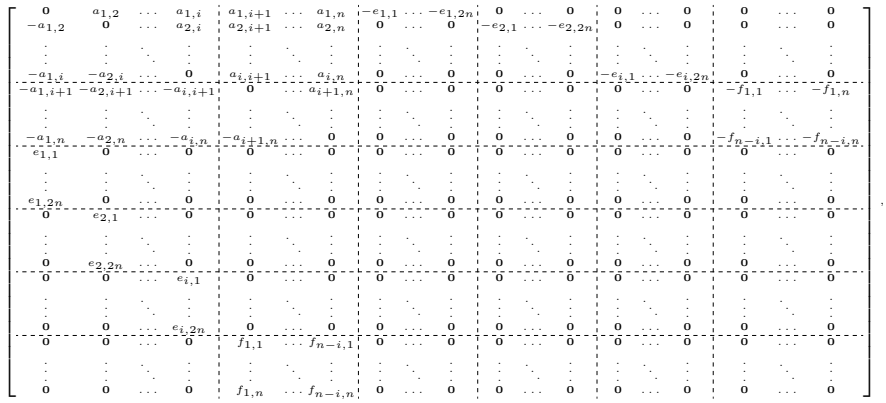
TheoremA'. Given an oracle deciding Alternating Matrix Space Isometry, the task of finding an isometry between two alternating matrix spaces $\mathcal{A}, \mathcal{B} \in \Lambda(n, \mathbb{F}_q)$, if it exists, can be solved using at most $q^{O(n)}$ oracle queries each of size at most $O(n^2)$, and in time either $q^{O(n)} \cdot n! = q^{\widetilde{O}(n)}$, or $q^{O(n+m)}$.

Proof. We first present the gadget construction. Then based on this gadget, we present the search-to-decision reduction.

Gadget construction. Let $\mathbf{A} = (A_1, \ldots, A_m)$ be an ordered linear basis of $\mathcal{A}$, and let $\mathtt{A} \in \mathrm{T}(n \times n \times m, \mathbb{F}_q)$ be the 3-way array constructed from $\mathbf{A}$, so we can write

$$\mathtt{A} = \begin{bmatrix} 0 & a_{1,2} & a_{1,3} & \cdots & a_{1,n} \\ -a_{1,2} & 0 & a_{2,3} & \cdots & a_{2,n} \\ -a_{1,3} & -a_{2,3} & 0 & \cdots & a_{3,n} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ -a_{1,n} & -a_{2,n} & -a_{3,n} & \cdots & 0 \end{bmatrix},$$

where $a_{i,j} \in \mathbb{F}^m$, $1 \leq i < j \leq n$ thought of as a vector coming out of the page. We first consider a 3-tensor $\widetilde{\mathtt{A}}_i$ constructed from $\mathtt{A}$, for any $1 \leq i \leq n-1$, as $\widetilde{\mathtt{A}}_i =$



where $e_{j,k}$ is the $(m + 2n(j-1) + k)$th standard basis vector, and $f_{j,k}$ is the $(m + 2ni + n(j-1) + k)$th standard basis vector. A pictorial description can be seen by combining Figure2 (for the $e_{j,k}$) and [32, Figure3] (for the $f_{j,k}$). We claim the following.

Claim10. If there exist invertible matrices P and Q to satisfy $P^t\widetilde{\mathtt{A}}_i P = \widetilde{\mathtt{B}}_i^Q$, then P must be in the form $\begin{bmatrix} P_{1,1} & 0 & 0 \\ 0 & P_{2,2} & 0 \\ P_{3,1} & P_{3,2} & P_{3,3} \end{bmatrix}$, where $P_{1,1}$ is a monomial matrix of size $i \times i$, $P_{2,2}$ is of size $(n-i) \times (n-i)$, and $P_{3,3}$ is of size $(2ni+n) \times (2ni+n)$. Furthermore, there exist such P and Q if and only if $\mathcal{A}$ and $\mathcal{B}$ are isometric by a matrix of the form $\begin{bmatrix} P_{1,1} & 0 \\ 0 & P_{2,2} \end{bmatrix}$ where $P_{1,1}$ is a monomial matrix of size $i \times i$.

Proof. This claim is immediate by combining the arguments for the FGS gadget [28] as used in [32], and the monomial-restricting gadget introduced in Section4.1. We only outline the argument and point out some subtle issues here. First, observe that for the lateral slices of $\widetilde{\mathtt{A}}_i$:

- The first i lateral slices have rank in $[2n, 3n)$. Note that the rank is strictly less than $3n$ because some tube fibers (coming out of the page) are 0 in the upper-left $n \times n$ sub-array.
- The next $n-i$ lateral slices have rank in $[n, 2n)$.
- The remaining $2ni+n$ lateral slices have rank in $[1, n)$ (since $i \geq 1$.)

Because of the above, for P and Q to satisfy $P^t\widetilde{\mathtt{A}}_i P = \widetilde{\mathtt{B}}_i^Q$, P must be in the required form. It is the furthermore statement that requires certain care. The only if direction is straightforward: after observing that P has to be of the above form, we can easily verify that $\begin{bmatrix} P_{1,1} & 0 \\ 0 & P_{2,2} \end{bmatrix}$ is an isometry from $\mathcal{A}$ to $\mathcal{B}$. For the if direction, starting from $\begin{bmatrix} P_{1,1} & 0 \\ 0 & P_{2,2} \end{bmatrix}$ and $Q_{1,1} \in \mathrm{GL}(m, \mathbb{F})$, we need to design $P_{3,3} \in \mathrm{GL}(2ni+n, \mathbb{F})$ and $Q_{2,2} \in \mathrm{GL}(2ni+n(n-i), \mathbb{F})$ such that letting

$$P = \begin{bmatrix} P_{1,1} & 0 & 0 \\ 0 & P_{2,2} & 0 \\ 0 & 0 & P_{3,3} \end{bmatrix} \quad \text{and} \quad Q = \begin{bmatrix} Q_{1,1} & 0 \\ 0 & Q_{2,2} \end{bmatrix},$$

we have $P^t\widetilde{\mathtt{A}}_i P = \widetilde{\mathtt{B}}_i^Q$. This can be achieved by combining the arguments for the only if directions in the proofs of Lemma8 and [32, Proposition3.3]. □

The search-to-decision reduction. Given these preparations, we now present the search-to-decision reduction for Alternating Matrix Space Isometry. Recall that this requires us to use the decision oracle $\mathcal{O}$ to compute an explicit isometry transformation $P \in \mathrm{GL}(n, q)$, if $\mathcal{A}$ and $\mathcal{B}$ are indeed isometric. Think of P as sending the standard basis $(\vec{e}_1, \ldots, \vec{e}_n)$ to another basis $(v_1, \ldots, v_n)$, where $\vec{e}_i$ and v_i are in $\mathbb{F}_q^n$. In the first step, we guess v_1, the image of $\vec{e}_1$, and a complement subspace of $\langle v_1 \rangle$, at the cost of $q^{O(n)}$. For each such guess, let P_1 be the matrix which sends $\vec{e}_1 \mapsto v_1$ and sends $\langle \vec{e}_2, \ldots, \vec{e}_n \rangle$ to the chosen complementary subspace arbitrarily. We apply P_1 to $\mathtt{A}$, and still call the resulting 3-way array $\mathtt{A}$ in the following. Then construct $\widetilde{\mathtt{A}}_1$ and $\widetilde{\mathtt{B}}_1$, and feed these two instances to the oracle $\mathcal{O}$. Note that, since $P_{1,1}$ (using notation as above) must be monomial, any equivalence between $\widetilde{\mathtt{A}}_1$ and $\widetilde{\mathtt{B}}_1$ must preserve our choice of v_1 up to scale.

Thus, clearly, if $\mathcal{A}$ and $\mathcal{B}$ are indeed isometric and we guess the correct image of $\vec{e}_1$, then the oracle $\mathcal{O}$ will return yes (and conversely). In the second step, we guess v_2, the image of $\vec{e}_2$, and a complement subspace of $\langle v_2 \rangle$ within $\langle \vec{e}_2, \ldots, \vec{e}_n \rangle$, at the cost of $q^{O(n)}$. Note here that the previous step guarantees that there is an isometry respecting the direct sum decomposition $\langle v_1 \rangle \oplus \langle \vec{e}_2, \ldots, \vec{e}_n \rangle$, so we need only search for a complement of v_2 within $\langle \vec{e}_2, \ldots, \vec{e}_n \rangle$, and not a more general complement of $\langle v_1, v_2 \rangle$ in all of $\mathbb{F}_q^n$. This is crucial for the runtime, as at the $n/2$ step, the latter strategy would result in searching through $q^{\Theta(n^2)}$ possibilities. For each such guess, we apply the corresponding transformation to $\mathtt{A}$ (and again call the resulting 3-way array $\mathtt{A}$). Then construct $\widetilde{\mathtt{A}}_2$ and $\widetilde{\mathtt{B}}_2$, and feed these two instances to the oracle $\mathcal{O}$. Clearly, if $\mathcal{A}$ and $\mathcal{B}$ are indeed isometric and we guess the correct image of $\vec{e}_2$ (and $\vec{e}_1$ from the previous step), then the oracle $\mathcal{O}$ will return yes. However, there is a small caveat here, namely we may guess some image of $\vec{e}_2$, such that $\mathcal{A}$ and $\mathcal{B}$ are actually isometric by some matrix P of the form $\begin{bmatrix} P_{1,1} & 0 \\ 0 & P_{2,2} \end{bmatrix}$ where $P_{1,1}$ is a monomial matrix of size 2 (instead of the more desired diagonal matrix). But this is fine, as it still ensures $P_{1,1}$ to be monomial, which is the key property to keep. This means that our choices of $\{v_1, v_2\}$ is correct as a set up to scaling, so we proceed. In general, in the ith step, we maintain the property that $\mathcal{A}$ and $\mathcal{B}$ are isometric by some $P = \begin{bmatrix} P_{1,1} & 0 \\ 0 & P_{2,2} \end{bmatrix}$ where $P_{1,1}$ is a monomial matrix of size $(i-1) \times (i-1)$. We guess v_i, the image of $\vec{e}_i$ in $\langle \vec{e}_i, \ldots, \vec{e}_n \rangle$, and a complement subspace of $\langle v_i \rangle$ within $\langle \vec{e}_i, \ldots, \vec{e}_n \rangle$. This cost is $q^{O(n)}$. For each such guess, we apply the corresponding transformation to $\mathtt{A}$ (and call the resulting 3-way array $\mathtt{A}$). Then construct $\widetilde{\mathtt{A}}_i$ and $\widetilde{\mathtt{B}}_i$, and feed these two instances to the oracle $\mathcal{O}$. Once we guess correctly, we ensure that $\mathcal{A}$ and $\mathcal{B}$ are isometric by $P = \begin{bmatrix} P_{1,1} & 0 \\ 0 & P_{2,2} \end{bmatrix}$ where $P_{1,1}$ is a monomial matrix of size $i \times i$. So after the $(n-1)$th step, we know that $\mathcal{A}$ and $\mathcal{B}$ are isometric by a monomial transformation. As the number of all monomial transformations is $(q-1)^n \cdot n! \leq q^n \cdot 2^{n \log n} = q^{\widetilde{O}(n)}$, we can enumerate all monomial transformations and check correspondingly. This gives an algorithm in time $q^{\widetilde{O}(n)}$. By resorting to Proposition11 which solves Alternating Matrix Space Monomial Isometry in time $q^{O(n+m)}$, we have an algorithm in time $q^{O(n+m)}$. Note that all the instances we feed into the oracle $\mathcal{O}$ are of size $O(n^2)$. This concludes the proof. □

4.3 A simply-exponential algorithm for monomial isometry of alternating matrix spaces

We now state the algorithm for monomial isometry used in TheoremA'.

Proposition11. Let $\mathcal{A}, \mathcal{B} \leq \Lambda(n, q)$ be m-dimensional. Then there exists a $q^{O(n+m)}$-time algorithm that decides whether $\mathcal{A}$ and $\mathcal{B}$ are monomially isometric, and if so, computes an explicit monomial isometry.

Proof. Let $\mathcal{A}, \mathcal{B} \leq \Lambda(n, q)$ be two m-dimensional alternating matrix spaces. Clearly, by incurring a multiplicative factor of q^n, we can reduce to the problem of testing whether $\mathcal{A}$ and $\mathcal{B}$ are permutationally isometric, i.e. whether there

exists a permutation matrix $T \in \mathrm{GL}(n,q)$, such that $T^t \mathcal{A} T = \mathcal{B}$. We will solve this problem in time $2^{O(n)} \cdot q^{O(m)}$. This would give an algorithm with total running time $q^n \cdot 2^{O(n)} \cdot q^{O(m)} = q^{O(n+m)}$. The basic idea of the algorithm comes from Luks's dynamic programming technique for Hypergraph Isomorphism [53].

Reducing to a generalized linear code equivalence problem. Suppose $\mathcal{A} = \langle A_1, \ldots, A_m \rangle$, and $\mathcal{B} = \langle B_1, \ldots, B_m \rangle$. Let $\mathtt{A}$ and $\mathtt{B}$ be the $n \times n \times m$ 3-way arrays formed by the given bases of $\mathcal{A}$ and $\mathcal{B}$. For $S \subseteq [n]$ of size s, let $(A_i)_S$ be the submatrix of A_i with row and column indices in S. Then let $\mathtt{A}_S$ be the $s \times s \times m$ 3-way array formed by $((A_1)_S, \ldots, (A_m)_S)$. Similarly we can define $\mathtt{B}_S$ for $S \subseteq [n]$. For each $S \subseteq [n]$ of size s, let $\mathrm{Iso}(\mathtt{A}_{[s]}, \mathtt{B}_S)$ be the coset in $S_n \times \mathrm{GL}(m,q)$, such that $(A,B) \in S_n \times \mathrm{GL}(m,q)$ if and only if the natural action of (A,B) sends $\mathtt{A}_{[s]}$ to $\mathtt{B}_S$. Since all the matrices are alternating, their diagonal entries are zero, and thus $\mathtt{A}_{\{i\}}$ and $\mathtt{B}_{\{i\}}$ are both the $1 \times 1 \times m$ zero vector for any i. It follows that if $s = 1$ and $S = \{i\}$, $\mathrm{Iso}(\mathtt{A}_{[1]}, \mathtt{B}_S) = G \times \mathrm{GL}(m,q)$, where G is the coset of S_n consisting of permutations sending 1 to i. Suppose we have computed $\mathrm{Iso}(\mathtt{A}_{[s]}, \mathtt{B}_S)$ for all $s < t$. Fix $T \subseteq [n]$, $|T| = t$, and let us compute $\mathrm{Iso}(\mathtt{A}_{[t]}, \mathtt{B}_T)$. For any $(A,B) \in \mathrm{Iso}(\mathtt{A}_{[t]}, \mathtt{B}_T)$, A sends $[t-1]$ to some $T' \subseteq T$ of size $t-1$. So in this case, $(A,B) \in \mathrm{Iso}(\mathtt{A}_{[t-1]}, \mathtt{B}_{T'})$, which has been computed. Let $T \setminus T' = \{t'\}$. On the other hand, for $(A,B) \in \mathrm{Iso}(\mathtt{A}_{[t-1]}, \mathtt{B}_{T'})$ to be in $\mathrm{Iso}(\mathtt{A}_t, \mathtt{B}_T)$, (A,B) needs to send the tth horizontal slice of $\mathtt{A}_{[t]}$ to the t'th horizontal slice of $\mathtt{B}_T$. We first identify T' with $[t-1]$. We then note that every horizontal slice of $\mathtt{A}_{[t]}$ has a row of zeros. So the problem now becomes: given two $(t-1) \times m$ matrices P and Q over $\mathbb{F}_q$, decide whether P and Q are the same under $G \leq S_{t-1} \times \mathrm{GL}(m,q)$. (Note that $G = \mathrm{Iso}(\mathtt{A}_{[t-1]}, \mathtt{B}_{T'})$ from above.) Clearly, this is a generalization of the Linear Code Equivalence problem. Furthermore, if we could solve this problem in time $2^{O(n)} \cdot q^{O(m)}$, we would have achieved our original goal.

Solving the generalized linear code equivalence problem. We solve the above problem again by a dynamic programming scheme as follows. For $R \subseteq [t-1]$ of size r, P_R denotes the $r \times m$ submatrix of P with row indices from R. Let $\mathrm{Iso}'(P_{[r]}, Q_R)$ be the coset in $S_{t-1} \times \mathrm{GL}(m,q)$, such that $(C,D) \in \mathrm{Iso}'(P_{[r]}, Q_R)$ if and only if the natural action of (C,D) sends $P_{[r]}$ to Q_R. If $r = 0$, then $\mathrm{Iso}'(P_\emptyset, Q_\emptyset) = G$ where $G \leq S_{t-1} \times \mathrm{GL}(m,q)$ is given as an input. Suppose we have computed $\mathrm{Iso}'(P_{[r]}, Q_R)$ for any $r < u$. Fix $U \subseteq [t-1]$, $|U| = u$, and let us compute $\mathrm{Iso}'(P_{[u]}, Q_U)$. For any $(C,D) \in \mathrm{Iso}'(P_{[u]}, Q_U)$, C sends $[u-1]$ to some $U' \subseteq U$ of size $u-1$. So in this case, $(A,B) \in \mathrm{Iso}(P_{[u-1]}, Q_U)$, which has been computed. Let $U \setminus U' = \{u'\}$. On the other hand, for $(C,D) \in \mathrm{Iso}(P_{[u-1]}, Q_{U'})$ to be in $\mathrm{Iso}(P_{[u]}, Q_U)$, D needs to send the uth row of $P_{[u]}$ to the u'th row of Q_U. This subcoset of $\mathrm{Iso}(P_{[u-1]}, Q_{U'})$ can be computed in time $q^{O(m)}$, by treating $\mathrm{GL}(m,q)$ as a permutation group on $\mathbb{F}_q^m$. We then take a union over size-$(u-1)$ subsets U' to obtain a generating set for $\mathrm{Iso}(P_{[u]}, Q_U)$. If necessary, we can reduce the generating set size by applying the standard permutation group machinery, as our time bound is $2^{O(n)} \cdot q^{O(m)}$, which is quite generous. □

6.2 Constructive Baer Correspondence and Theorems A and B

Let us review Baer's Correspondence [7], which connects alternating bilinear maps with p-groups of class 2 and exponent p. Let P be a p-group of class 2 and exponent p, $p > 2$. Suppose the commutator subgroup $[P, P] \cong \mathbb{Z}_p^m$ and $P/[P, P] \cong \mathbb{Z}_p^n$. Then the commutator map $[,] : P/[P, P] \times P/[P, P] \to [P, P]$ is an alternating bilinear map. Conversely, let $\phi : \mathbb{Z}_p^n \times \mathbb{Z}_p^n \to \mathbb{Z}_p^m$ be an alternating bilinear map. Then a p-group of class 2 and exponent p, denoted as P_ϕ can be defined as follows. The group elements are from $\mathbb{Z}_p^n \times \mathbb{Z}_p^m$, and the group product $\cdot$ is defined as

$$(u, v) \cdot (u', v') = (u + u', v + v' + \tfrac{1}{2}\phi(u, u')).$$

We say that $(A, B) \in \mathrm{GL}(n, p) \times \mathrm{GL}(m, p)$ is a pseudo-autometry of ϕ, if $\phi = B \circ \phi \circ A$. Wilson [71] elucidated the structure of $\mathrm{Aut}(P_\phi)$ in terms of the pseudo-autometry group of ϕ, that we denote $\Psi\,\mathrm{Aut}(\phi)$. Here we recall the consequence of Wilson's result that we need for counting group isomorphisms.

Proposition23 (Wilson [71, Prop.3.8], see [15, Prop.2.4] for notation closer to ours). For $\phi : \mathbb{Z}_p^n \times \mathbb{Z}_p^n \to \mathbb{Z}_p^m$ an alternating bilinear map,

$$|\,\mathrm{Aut}(P_\phi)| = |\Psi\,\mathrm{Aut}(\phi)|p^{nm},$$

where $\Psi\,\mathrm{Aut}(\phi)$ denotes the pseudo-autometry group of ϕ.

We then state a lemma which can be viewed as a constructive version of Baer's Correspondence, communicated to us by James B. Wilson.

Lemma24 (Constructive version of Baer's Correspondence for matrix groups). Let p be an odd prime. Over the finite field $\mathbb{F} = \mathbb{F}_{p^e}$, Alternating Matrix Space Isometry is equivalent to Group Isomorphism for matrix groups over $\mathbb{F}$ that are p-groups of class 2 and exponent p. More precisely, there are functions computable in time $\mathrm{poly}(n, m, \log|\mathbb{F}|)$:

- $G : \Lambda(n, \mathbb{F})^m \to \mathrm{M}(n + m + 1, \mathbb{F})^{n+m}$ and
- $\mathrm{Alt} : \mathrm{M}(n, \mathbb{F})^m \to \Lambda(m, \mathbb{F})^{O(m^2)}$

such that: (1) for an alternating bilinear map $\mathbf{A}$, the group generated by $G(\mathbf{A})$ is the Baer group corresponding to $\mathbf{A}$, (2) G and Alt are mutually inverse, in the sense that the group generated by $G(\mathrm{Alt}(M_1, \ldots, M_m))$ is isomorphic to the group generated by $M_1, \ldots, M_m$, and conversely $\mathrm{Alt}(G(\mathbf{A}))$ is pseudo-isometric to $\mathbf{A}$.

Proof. First, let G be a p-group of class 2 and exponent p given by m generating matrices of size $n \times n$ over $\mathbb{F}$. Then from the generating matrices of G, we first compute a generating set of $[G, G]$, by just computing all the commutators of the given generators. We can then remove those redundant elements from this generating set in time $\mathrm{poly}(\log|[G, G]|, \log|\mathbb{F}|)$, using Luks' result on computing with solvable matrix groups [51]. We then compute a set of representatives of a non-redundant generating set of $G/[G, G]$, again using Luks's aforementioned

result. From these data we can compute an alternating bilinear map representing the commutator map of G in time $\mathrm{poly}(n, m, \log|\mathbb{F}|)$. Conversely, let an alternating bilinear map be given by $\mathbf{A} = (A_1, \ldots, A_m) \in \Lambda(n, \mathbb{F})^m$. From $\mathbf{A}$, for $i \in [n]$, construct $B_i = [A_1 e_i, \ldots, A_m e_i] \in \mathrm{M}(n \times m, \mathbb{F})$, where e_i is the ith standard basis vector of $\mathbb{F}^n$. That is, the jth column of B_i is the ith column of A_j. Then for $i \in [n]$, construct

$$\widetilde{B}_i = \begin{bmatrix} 1 & e_i^t & 0 \\ 0 & I_n & B_i \\ 0 & 0 & I_m \end{bmatrix} \in \mathrm{GL}(1+n+m, \mathbb{F}),$$

where $e_i \in \mathbb{F}^n$, and for $j \in [m]$, construct

$$\widetilde{C}_j = \begin{bmatrix} 1 & 0 & e_j^t \\ 0 & I_n & 0 \\ 0 & 0 & I_m \end{bmatrix} \in \mathrm{GL}(1+n+m, \mathbb{F}),$$

where $e_j \in \mathbb{F}^m$. Let $G(\mathbf{A})$ be the matrix group generated by $\widetilde{B}_i$ and $\widetilde{C}_j$. Then it can be verified easily that, $G(\mathbf{A})$ is isomorphic to the Baer group corresponding to the alternating bilinear map defined by $\mathbf{A}$. In particular, $[G, G] \cong \mathbb{F}^m \cong \mathbb{Z}_p^{em}$ (isomorphism of abelian groups), and $G/[G, G] \cong \mathbb{F}^n \cong \mathbb{Z}_p^{en}$. This construction can be done in time $\mathrm{poly}(n, m, \log|\mathbb{F}|)$. □

Proof of TheoremA. The search-to-decision reduction follows from TheoremA', using the $q^{O(n+m)}$-time algorithm, with the constructive version of Baer's Correspondence in the model of matrix groups over finite fields (Lemma24). In more detail, given Lemma24 we can follow the procedure in the proof of TheoremA'. For the given p-groups, we compute their commutator maps. Then whenever we need to feed the decision oracle, we transform from the alternating bilinear map to a generating set of a p-group of class 2 and exponent p with this bilinear map as the commutator map. After getting the desired pseudo-isometry for the alternating bilinear maps, we can easily recover an isomorphism between the originally given p-groups. □

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